

An Overview of Force Measurement

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Abstract—The abstract goes here.

Index Terms—Interaction Force, Force Measurement.

I. INTRODUCTION

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II. FORCE SENSING METHODS

There are many methods to measure forces exerted on objects. These methods differ from one another in the physical principle of sensing. In this section, we will explain some of these methods and compare between advantages and disadvantages of each methodes, as shown in Table (??).

A. Resistive force transducer

B. Capacitive force transducer

C. Inductive force transducer

D. Piezoelectric force transducer

In piezoelectric force sensors, the measuring element is based on a crystal (quartz or ceramic) as illustrated in Fig.(1)(b). The crystal produces an electrical charge when a mechanical force is applied to it. The amount of electric charge generated in the crystal is proportional to the amplitude of the applied force. Most piezoelectric force sensors operate with a measuring element. It consists of thin crystal disks. Depending on whether the sensor is to measure compression or shear forces, the disks are cut from the crystal with different cutting angles[7]. Two primary circuits are used to process the output signal of the piezoelectric transducer. They are the charge amplifier circuit and the voltage amplifier circuit. A charge amplifier as illustrated in Figure.(1)(a) is used to amplify the electric charge generated by the crystal and the voltage resulting from the application of mechanical force to the piezoelectric element is given by (1)

$$V_{out} = \frac{-q_{in}}{C_f} \quad (1)$$

where c_f is the feedback capacitor, q_{in} the amount of charge generated. Piezoelectric materials possess a high modulus of elasticity, which gives piezoelectric force sensors a wide dynamic measurement range [1]. Piezoelectric force sensors can

measure forces ranging from the millinewton to the meganewton level. They also have the capability to amplify the value up to 105 [2]. The equivalent of the sensitivity factor is the piezoelectric coefficient d , which depends only on the material used [2]. It is a measure of the amount of charge caused by applied force expressed in pC/N (2.3 for X-cut quartz plate, about 30 for polypropylene ferroelectrets and more than 200 for lead zirconium titanate PZT) [2]. To produce sensors with higher sensitivity, several crystal disks can be stacked on top of one another and connected electrically in parallel. Alternatively, a piezoelectric material with higher sensitivity can be used (e.g. PiezoStar crystals) [7]. In the literature, Krajewski et al.[3] designed flexible two-dimensional sensor by integrating conductive fibrous electrodes and polyvinylidene fluoride (PVDF) fibers into a conventional plain woven polyester fabric. The electrical signal generated by the sensors was correlated to the fineness of the PVDF fibers used, the distance between the electrodes and the nature of the electrodes. Relationships between sensor output signal under load and the type of structure were thus established. Kim et al.[4] discussed the detailed mechanism of these piezoelectric-type, static force sensing methods that use more than the direct piezoelectric effect. They also highlighted the challenges and potentials of each method for static force sensing applications. Yu et al.[5] proposed a new flexible piezoelectric tactile sensor array based on polyvinylidene fluoride (PVDF) film for measuring three-axis dynamic contact force distribution. The sensor array can be easily integrated onto a curved surface, such as robotic and prosthetic hands, due to its excellent flexibility.

E. Fiber optic force transducer

F. Vibrating wire force transducer

III. STATIC VS DYNAMIC FORCE MEASUREMENT

IV. MULTI-DIMENSIONAL FORCE MEASUREMENT

V. FORCE ESTIMATION

VI. FORCE MEASUREMENT APPLICATIONS

A. Force measurement in monitoring and testing tasks

B. Use of force measurement in automated systems

C. Force measurement in research and development laboratories

VII. CONCLUSION

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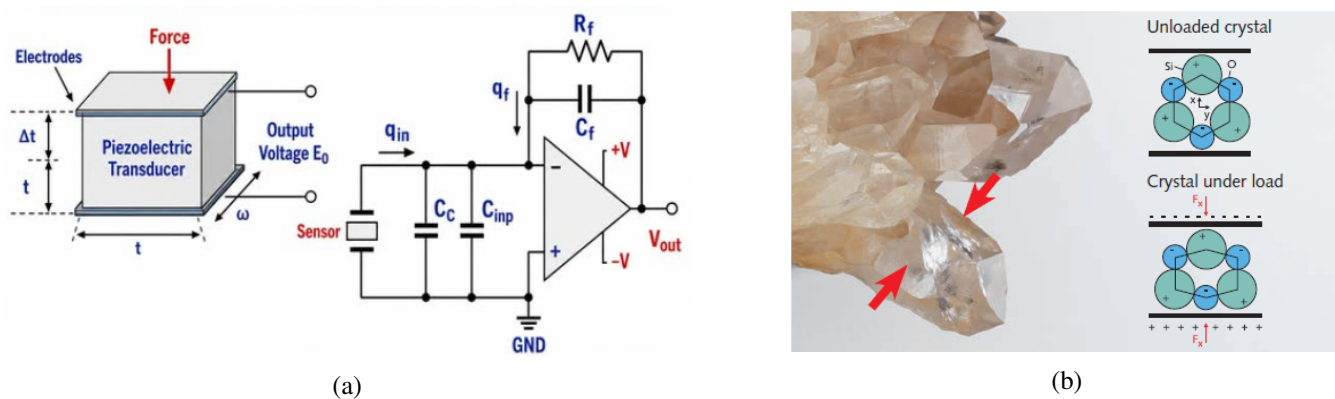


Fig. 1: (a) Piezo sensor and charge amplifier circuit [6]. (b) The mechanical load on a crystal produces an electrical charge [7]

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